OPERATING SYSTEMS • CSA0411

REAL-TIME OPERATING SYSTEM IN TRAFFIC SIGNAL CONTROL

Technical Presentation



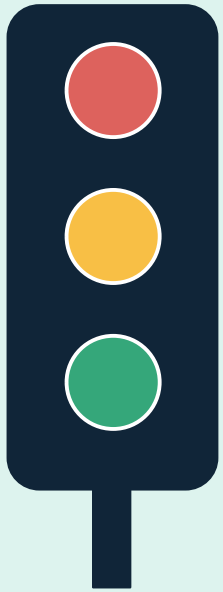
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Introduction and Objectives

Why timing matters in everyday traffic management

1. A traffic signal controls vehicle movement at a road junction.
2. An RTOS performs important tasks within fixed time limits.
3. It reads inputs from timers, sensors and control switches.
4. It changes signal lights in a planned and safe sequence.
5. Objective: give timely and predictable signal control.

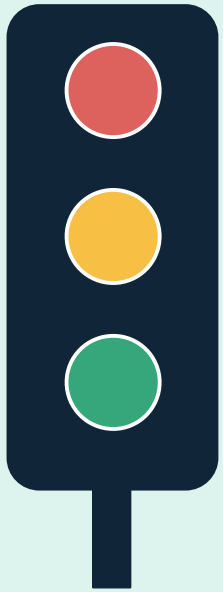


TIME-CRITICAL CONTROL

Problem Statement

The challenge of controlling a busy intersection

1. Traffic flow changes throughout the day.
2. Late signal changes may cause confusion and congestion.
3. Multiple inputs must be handled without missing deadlines.
4. A normal OS may not guarantee the required response time.
5. Solution: use a real-time OS to schedule critical tasks.

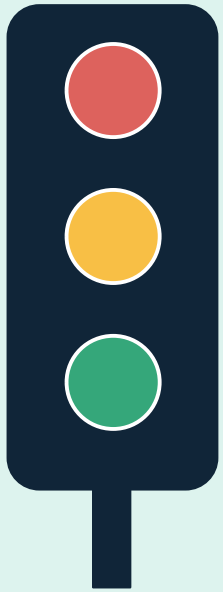


THE TIMING CHALLENGE

Type of OS and Justification

A real-time OS is chosen for predictable response

1. A Real-Time Operating System (RTOS) manages timed tasks.
2. A hard real-time design treats missed critical deadlines as failures.
3. Priority scheduling runs urgent tasks before less urgent tasks.
4. Timers help each signal phase run for its set duration.
5. Predictable response makes RTOS suitable for signal control.



WHY RTOS?

OS Structure and Components

A simple view of how the controller works

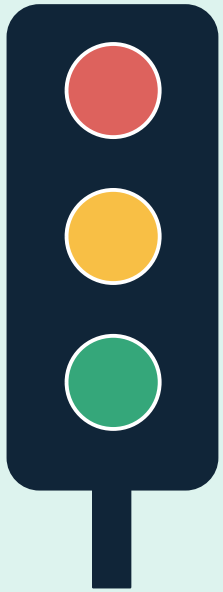


1. Sensor input reports traffic or pedestrian requests.
2. The kernel coordinates tasks and hardware resources.
3. The scheduler selects the highest-priority ready task.
4. Timers trigger the next traffic-light phase.
5. Output control turns the correct signal lamps on.

System Calls and Timing

How application tasks request operating-system services

1. Task creation starts the signal-control process.
2. Delay or sleep calls wait for the next timed phase.
3. Timer calls track durations and activate timed events.
4. Semaphore calls coordinate sensor and controller tasks.
5. Device I/O calls read inputs and update signal outputs.



TASKS + TIMERS

Advantages, Limitations and Conclusion

Key points to remember before presenting

ADVANTAGES

- Predictable response to events
- Priority-based task execution
- Accurate signal phase timing

LIMITATIONS

- Requires careful task design
- Missed deadlines can affect safety

Conclusion: RTOS makes traffic-signal control timely, coordinated and predictable.

THANK YOU